difficult to extract and analyze, theorists have based much of their research on approximate equations that are far easier to work with. While there are good reasons to believe that the approximate equations should, in many circumstances, give answers close to those given by the true equations, approximations—like translations—always miss something. For this reason, certain key problems have proved beyond the approximate equations' mathematical reach, significantly impeding progress.

For the imprecision inherent in textual translations, readers have a couple of immediate remedies. The best option, if the reader's linguistic skills are up to the task, is to consult the original manuscript. At the moment, the analog of this option is not available to string theorists. By virtue of the consistency of the dictionary developed by Witten and others, we have strong evidence that all five string theories are different descriptions of a single master theory, M-theory, but researchers have yet to develop a complete understanding of this theoretical nexus. We have learned much about M-theory in the last few years, but we still have far to go before anyone could sensibly claim that it is properly or completely understood. In string theory, it's as if we have five translations of a yet-to-be-discovered master text.

Another helpful remedy, well known to readers of translations who either don't have the original (as in string theory) or, more commonly, don't understand the language in which it's written, is to consult several translations of the master text into languages with which they are familiar. Passages on which the translations agree give confidence; passages on which they differ flag possible inaccuracies or highlight different interpretations. It is this approach that Witten made available with his discovery that the five string theories are different translations of the same underlying theory. In fact, his discovery provided an extremely powerful version of this line of attack that is best understood through a slight extension of the translation analogy.

Imagine a master manuscript infused with such an enormous range of puns, rhymes, and offbeat, culture-sensitive jokes, that the complete text cannot be expressed gracefully in any single one of five given languages into which it is being translated. Some passages might translate into Swahili with ease, while other portions might prove thoroughly